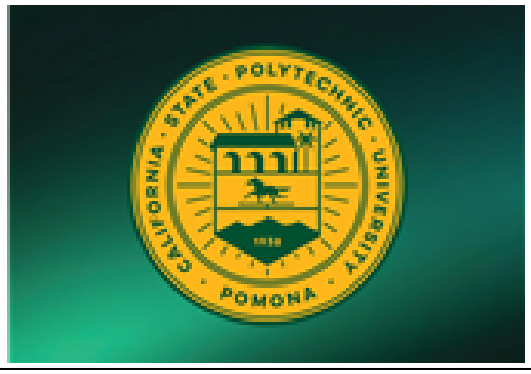
# Project Name: ControlShield



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# DESCRIPTION

ECE4311 – Network Forensics: Development of a PLC Guardian HMI for the M221 PLC on an HMIS5T panel, focusing on cybersecurity and network forensics for industrial control systems.

# BILL OF MATERIAL

## Controller



| | |
|---|---|
| **Reference** | TM221CE24T |
| **Description** | TM221CE24T (screw)<br>14 digital inputs, 10 source transistor outputs (0,5 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 24 Vdc power supply controller with removable terminal blocks. |
| **Power supplied to the IO bus** | 5V: 520 mA / 24V: 200 mA |

# HARDWARE CONFIGURATION

## MyPLC - TM221CE24T

### Digital Inputs

| Used | Address | Filtering | Latch | Run/Stop | Events | Priority | Subroutine |
|---|---|---|---|---|---|---|---|
| | %I0.0 | 3 ms | | | Not Used | | |
| | %I0.1 | 3 ms | | | Not Used | | |
| | %I0.2 | 3 ms | | | Not Used | | |
| | %I0.3 | 3 ms | | | Not Used | | |
| | %I0.4 | 3 ms | | | Not Used | | |
| | %I0.5 | 3 ms | | | Not Used | | |
| | %I0.6 | 3 ms | | | Not Used | | |
| | %I0.7 | 3 ms | | | Not Used | | |
| | %I0.8 | 3 ms | | | Not Used | | |
| | %I0.9 | 3 ms | | | Not Used | | |
| | %I0.10 | 3 ms | | | Not Used | | |
| | %I0.11 | 3 ms | | | Not Used | | |
| | %I0.12 | 3 ms | | | Not Used | | |
| | %I0.13 | 3 ms | | | Not Used | | |

### Digital Outputs

| Used | Address | Status Alarm | Fallback value | Used by |
|---|---|---|---|---|
| X | %Q0.0 | | 0 | User logic |
| X | %Q0.1 | | 0 | User logic |
| X | %Q0.2 | | 0 | User logic |
| | %Q0.3 | | 0 | |
| | %Q0.4 | | 0 | |
| | %Q0.5 | | 0 | |
| | %Q0.6 | | 0 | |
| | %Q0.7 | | 0 | |
| | %Q0.8 | | 0 | |
| | %Q0.9 | | 0 | |

## Analog Inputs

| Used | Address | Type | Scope | Range | Filter | Sampling |
|---|---|---|---|---|---|---|
| | %IW0.0 | 0 - 10 V | Normal | 0-1000 | 0 | |
| | %IW0.1 | 0 - 10 V | Normal | 0-1000 | 0 | |

## Fast Counters

| Used | Address | Input | Configured | Preset | Double Word |
|---|---|---|---|---|---|
| | %FC0 | %I0.2 | NotUsed | 0 | |
| | %FC1 | %I0.3 | NotUsed | 0 | |
| | %FC2 | %I0.4 | NotUsed | 0 | |
| | %FC3 | %I0.5 | NotUsed | 0 | |

## High Speed Counters

| Used | Address | Type |
|---|---|---|
| | %HSC0 | Not Configured |
| | %HSC1 | Not Configured |
| | %HSC2 | Not Configured |
| | %HSC3 | Not Configured |

## Pulse Generators

| Configured | Address | Type |
|---|---|---|
| | %PLS0/%PWM0/%PTO0/%FREQGEN0 | Not Configured |
| | %PLS1/%PWM1/%PTO1/%FREQGEN1 | Not Configured |

## ETH1

| | |
|---|---|
| **Device name:** | M221 |
| **IP Mode:** | Fixed |
| **IP address:** | 192.168.8.11 |
| **Subnet mask:** | 255.255.255.0 |
| **Gateway address:** | 0.0.0.0 |
| **Transfer Rate:** | Auto |
| **Security Parameters:** | Programming protocol enabled |
| | Auto discovery protocol enabled |
| | Modbus server enabled |
| | EtherNet/IP protocol enabled |

# SL1 (Serial line)

## Physical Settings

| | |
|---|---|
| **Device:** | None |
| **Baud rate:** | 19200 |
| **Parity:** | Even |
| **Data bits:** | 8 |
| **Stop bits:** | 1 |
| **Physical medium:** | RS-485 |
| **Polarization:** | No |

## Protocol Settings

| | |
|---|---|
| **Protocol:** | Modbus |
| **Response timeout (× 100 ms):** | 10 |
| **Time between frames (ms):** | 10 |
| **Transmission mode:** | RTU |
| **Addressing:** | Slave |
| **Address:** | 1 |

# SOFTWARE CONFIGURATION

## Constant Words

### KW

**Allocation:** Automatic

**Allocated:** 0

| Used | Equ Used | Address | Symbol | Value | Comment |
|---|---|---|---|---|---|

### KD

**Allocation:** Automatic

**Allocated:** 0

| Used | Equ Used | Address | Symbol | Value | Comment |
|---|---|---|---|---|---|

### KF

**Allocation:** Automatic

**Allocated:** 0

| Used | Equ Used | Address | Symbol | Value | Comment |
|---|---|---|---|---|---|

# Network Objects

## Input Assembly (Ethernet/Ip)

| Used | Address | Symbol | Fallback value | Comment |
|---|---|---|---|---|

## Output Assembly (Ethernet/Ip)

| Used | Address | Symbol | Comment |
|---|---|---|---|

## Input Registers (Modbus Tcp)

| Used | Address | Symbol | Fallback value | Comment |
|---|---|---|---|---|

## Output Registers (Modbus Tcp)

| Used | Address | Symbol | Comment |
|---|---|---|---|

## Digital inputs (IOScanner)

| Used | Address | Channel | Symbol | Comment |
|---|---|---|---|---|

## Digital outputs (IOScanner)

| Used | Address | Channel | Fallback value | Symbol | Comment |
|---|---|---|---|---|---|

## Input registers (IOScanner)

| Used | Address | Channel | Symbol | Comment |
|---|---|---|---|---|

## Output registers (IOScanner)

| Used | Address | Channel | Fallback value | Symbol | Comment |
|---|---|---|---|---|---|

# Software Objects

## Timers

**Allocation:** Automatic

**Allocated:** 2

| Used | Address | Symbol | Type | Retentive | Dynamic Preset | Time Base | Preset | Comment |
|---|---|---|---|---|---|---|---|---|
| X | %TM0 | TICK | TON | | | 1 ms | 100 | |
| X | %TM1 | TOCK | TOF | | | 1 ms | 100 | |

## Counters

**Allocation:** Automatic

**Allocated:** 1

| Used | Address | Symbol | Preset | Comment |
|---|---|---|---|---|
| X | %C0 | TANK_N | 80 | |

## LIFO/FIFO Registers

**Allocation:** Automatic

**Allocated:** 0

## Drums

**Allocation:** Automatic

**Allocated:** 0

## Shift Bit Registers

**Allocation:** Automatic

**Allocated:** 0

## Step Counters

**Allocation:** Automatic

**Allocated:** 0

## Schedule Blocks

**Allocation:** Automatic

**Allocated:** 0

## RTC



CPP

## PID

| Used | PID | Symbol | Type | Comment |
|---|---|---|---|---|

## Grafcet Steps

**Allocation:** Automatic

**Allocated:** 0

# PROGRAM

## Behavior

| | |
|---|---|
| **Functional level:** | Level 14.0 |
| **Starting mode:** | Start In Previous State |
| **Watchdog:** | 250 ms |
| **Fallback behavior:** | Fallback value |
| **String end character:** | CR (Carriage Return) |

## Memory consumption

A successful compilation is required to obtain memory information.

## Application Architecture

### Master Task

| | |
|---|---|
| **Scan mode:** | Normal |
| **POU list:** | 1 - Main |

### Periodic Task

| | |
|---|---|
| **Period:** | 255 ms |

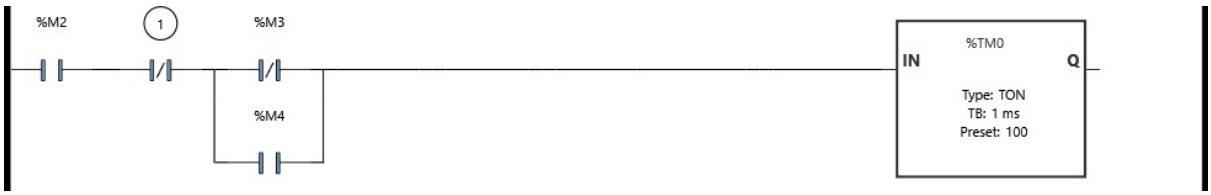
# POU

## Master Task

1 - Main

Master Task

*Rung0 - Start/Stop*



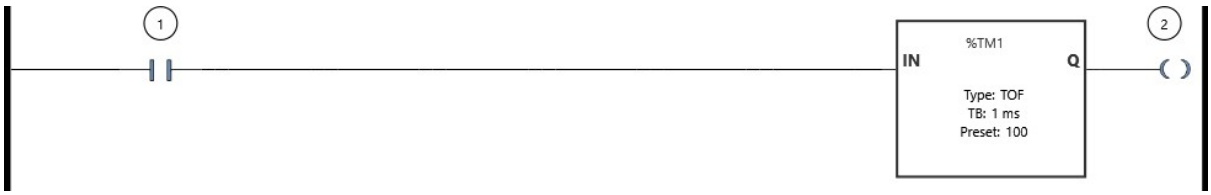

Legend:

1 %TM1.Q

Variables used:

| | |
|---|---|
| %M2 | STARTBUTTON |
| %M3 | STOP |
| %M4 | MAINT_BYPASS |
| %TM0 | TICK |
| %TM1.Q | TOCK.Q |

*Rung1 - Simulated Encoder*




Legend:

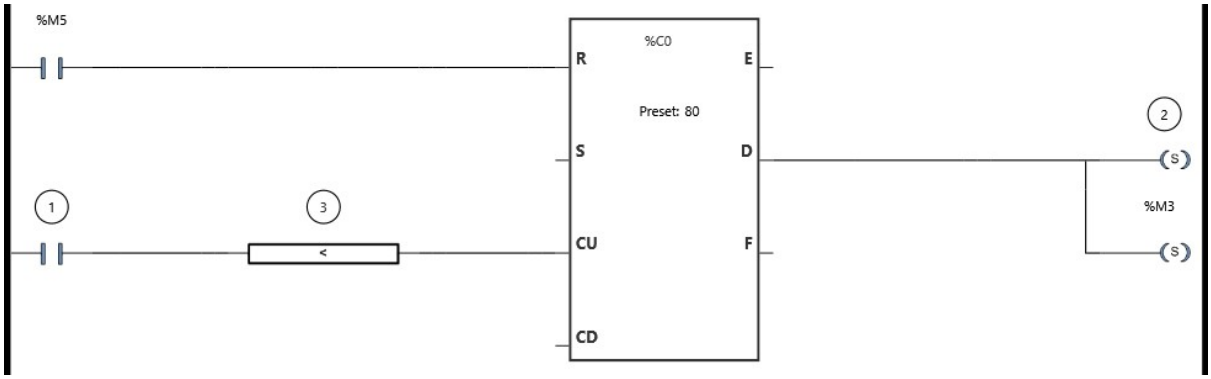
1 %TM0.Q

2 %Q0.0

Variables used:

| | |
|---|---|
| %Q0.0 | OUTPUT_1 |
| %TM0.Q | TICK.Q |
| %TM1 | TOCK |

*Rung2 - Tank Logic*



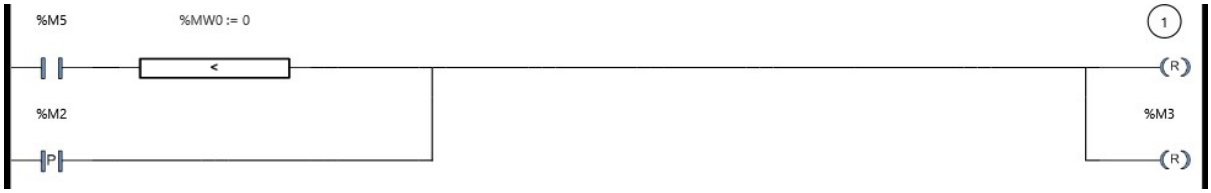

Legend:

1 %Q0.0
2 %Q0.1
3 %MW0 := %C0.V

Variables used:

| | |
|---|---|
| %C0 | TANK_N |
| %C0.V | TANK_N.V |
| %M3 | STOP |
| %M5 | RESET |
| %MW0 | TANK_VALUE |
| %Q0.0 | OUTPUT_1 |
| %Q0.1 | OUTPUT_2 |

*Rung3 - Reset Logic*




Legend:

1 %Q0.1

Variables used:

| | |
|---|---|
| %M2 | STARTBUTTON |
| %M3 | STOP |
| %M5 | RESET |
| %MW0 | TANK_VALUE |
| %Q0.1 | OUTPUT_2 |

*Rung4 - Tank Overload Flag*




Legend:

1 %Q0.2

Variables used:

| | |
|---|---|
| %C0.V | TANK_N.V |
| %Q0.2 | TANK_OVERLOAD |

# SYMBOLS

| Used | Address | Symbol | Comment |
|---|---|---|---|
| X | %C0 | TANK_N | |
| X | %M2 | STARTBUTTON | |
| X | %M3 | STOP | |
| X | %M4 | MAINT_BYPASS | |
| X | %M5 | RESET | |
| X | %MW0 | TANK_VALUE | |
| X | %Q0.0 | OUTPUT_1 | |
| X | %Q0.1 | OUTPUT_2 | |
| X | %Q0.2 | TANK_OVERLOAD | |
| X | %TM0 | TICK | |
| X | %TM1 | TOCK | |

# CROSS-REFERENCE TABLE

| Address | Object | Rung | Code |
|---|---|---|---|
| %C0... | 1 - Main | Rung2 - Tank Logic | %C0 |
| %C0.V. | 1 - Main | Rung2 - Tank Logic | --[<]-- %MW0 := %C0.V |
| | | Rung4 - Tank Overload Flag | --[<]-- %C0.V > 80 |
| %M2... | 1 - Main | Rung0 - Start/Stop | --\| \|-- |
| | | Rung3 - Reset Logic | --\|P\|-- |
| %M3... | 1 - Main | Rung0 - Start/Stop | --\|/\|-- |
| | | Rung2 - Tank Logic | --(S)-- |
| | | Rung3 - Reset Logic | --(R)-- |
| %M4... | 1 - Main | Rung0 - Start/Stop | --\| \|-- |
| %M5... | 1 - Main | Rung2 - Tank Logic | --\| \|-- |
| | | Rung3 - Reset Logic | --\| \|-- |
| %MW0.. | 1 - Main | Rung2 - Tank Logic | --[<]-- %MW0 := %C0.V |
| | | Rung3 - Reset Logic | --[<]-- %MW0 := 0 |
| %Q0.0. | 1 - Main | Rung1 - Simulated Encoder | --( )-- |
| | | Rung2 - Tank Logic | --\| \|-- |
| %Q0.1. | 1 - Main | Rung2 - Tank Logic | --(S)-- |
| | | Rung3 - Reset Logic | --(R)-- |
| %Q0.2. | 1 - Main | Rung4 - Tank Overload Flag | --( )-- |
| %TM0.. | 1 - Main | Rung0 - Start/Stop | %TM0 |
| %TM0.Q | 1 - Main | Rung1 - Simulated Encoder | --\| \|-- |
| %TM1.. | 1 - Main | Rung1 - Simulated Encoder | %TM1 |
| %TM1.Q | 1 - Main | Rung0 - Start/Stop | --\|/\|-- |